

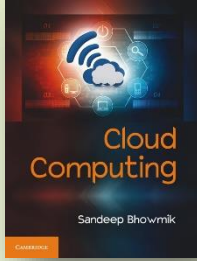
Cloud Computing

Sandeep Bhowmik

Chapter 9

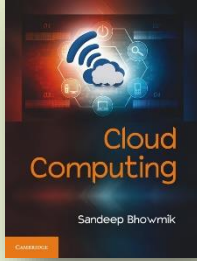
Scaling in the Cloud

Cambridge University Press



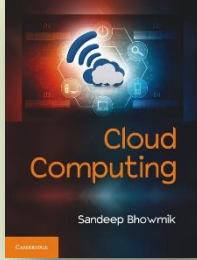
Scaling in the Cloud

- Resource virtualization technique creates room for the adoption of dynamic approach in computing resource provisioning.
- The dynamic resource provisioning approach in turn creates scope for developing scalable computing systems and applications.



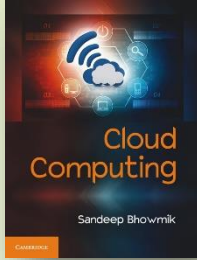
What is Scaling

- The ability of a system to grow and shrink as per workload is known as scaling.
- In computing, scaling represents the capability of a system or application to deal with varying workload efficiently.
- A system that scales well can maintain its level of performance or efficiency when works under larger or growing operational demands.



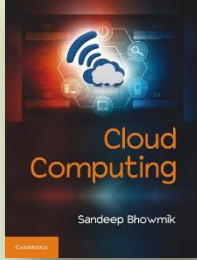
What is Scaling

- The *scalability* of a system is measured by the maximum workload it can competently handle at any particular moment.
- The point at which a system or application can no more handle additional workload efficiently, is known as its *limit of scalability*.
- Scalability reaches its limit when a system's architecture does not support scaling anymore, or some critical hardware resources run out.



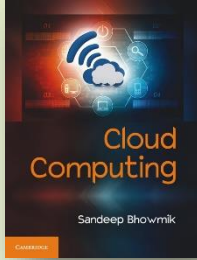
Scaling in Traditional Computing

- The infrastructure architecture is not dynamic in traditional computing system, which prevents implementation of *dynamic scaling*.
- The traditional *static scaling* approach require system restart which causes service disruption.
- In traditional computing, System designers always built computing systems by arranging resources to meet peak demand, wasting resources, and increasing cost.



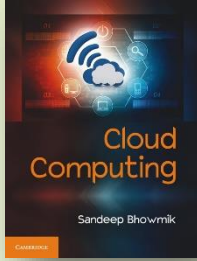
Scaling in Cloud Computing

- Scaling in cloud is dynamic in nature and apart from few special cases, automatic also.
- Dynamic scaling enables to keep a system perform consistently during massive demand by expanding it at the similar pace with the growing demand.
- The scalability in cloud is offered in a transparent manner.
- The automatic scaling ability provide the impression of infinite resources to its consumers.



Scaling in Cloud is Reversible

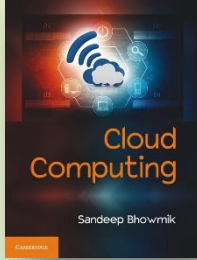
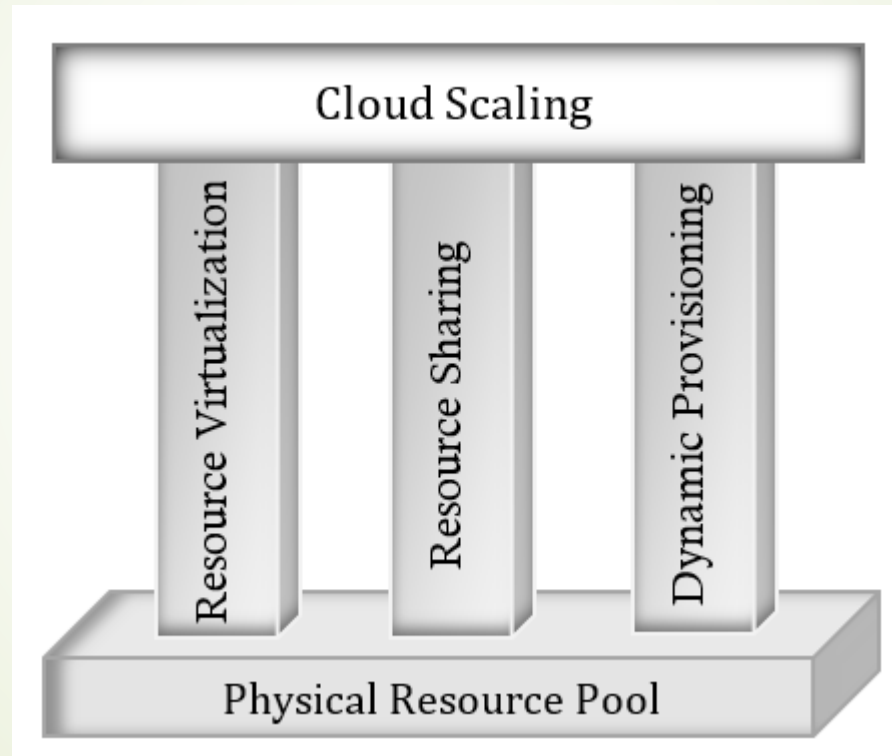
- In any computing environment, scaling-down is as important as scaling-up from business point of view.
- Reversible scaling is critical as system performance should not hamper while releasing resources.
- The presence of downward scaling capability has made cloud computing more attractive.
- Anytime during the running of a system, extra resource capacity can be given back to the resource pool.

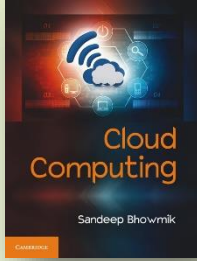


Foundation of Cloud Scaling

- The auto-scaling feature in cloud computing has been achieved based on three pillars –
 - Resource virtualization – eases resource management tasks and reduces complexity of system development.
 - Resource sharing – allows optimal resource utilization by sharing resources among multiple users as well as applications.
 - Dynamic resource provisioning – supply (or, reclaim) resources on-demand, in response to the increasing (or, decreasing) workload

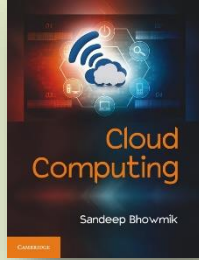
Foundation of Cloud Scaling





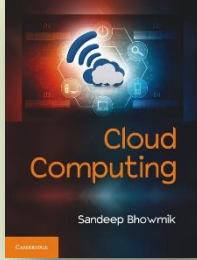
Foundation of Cloud Scaling

- A system's ability of scaling may depend on various parameters.
- Among them, two important stuffs which are dealt meticulously in the development of scalable cloud computing system are –
 - Capacity planning on regular basis
 - Load balancing



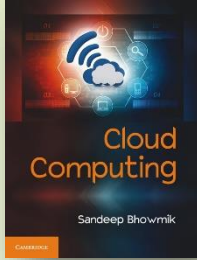
Scalable Application

- Application architecture should be well suited to scale in a scalable architectural environment.
- Application's scalability depends on its design, the types of algorithms applied and data structures used.
- It is not possible to take full advantage from a scalable computing infrastructure if application architecture is not scalable. Both have to work together to maximize the gain.



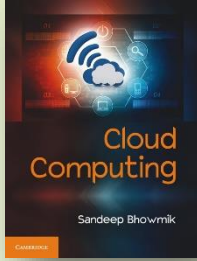
Scaling Strategies in Cloud

- In cloud computing, scaling strategy is dynamic.
- In the dynamic scaling, a system can be resized during its execution without restarting or interrupting any service.
- The critical task of dynamic capacity alteration can happen in two ways -
 - Manually – A system can be scaled while running, by executing commands through application interface.
 - Automatically – Implemented through program that can automatically adjust system capacity by observing the actual demand.



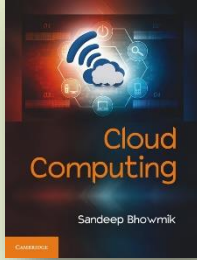
Scaling Strategies in Cloud

- Real power of cloud scaling lies in automatic scaling ability.
- The dynamic auto scaling is generally referred as *auto scaling*, which is also known as *cloud scaling*.
- Auto scaling can be implemented in two different ways –
 - Scaling based on a predefined schedule, known as *proactive scaling*.
 - Scaling based on current actual demand, known as *reactive scaling*.



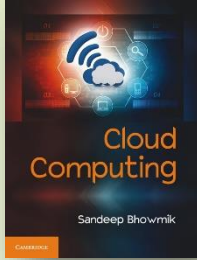
Proactive Scaling

- Proactive scaling strategy does not wait for demand to increase or decrease in expected circumstances.
- In cases where the expected increase or decrease of demand is known, a pre-programmed plan is placed to automatically alter resource capacity.
- Such scaling strategy, that does not wait for workload to change, rather alters capacity in advance based on a predefined schedule, is known as *proactive scaling*.



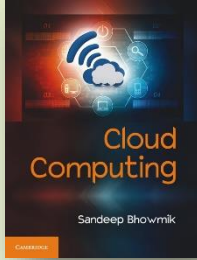
Proactive Scaling

- The proactive scaling schedules are implemented in two different ways -
 - Proactive cyclic scaling – This type of proactive scaling event takes place at fixed regular intervals.
 - Proactive event-based scaling – This is situation or event based.



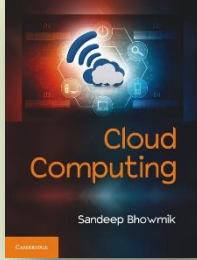
Reactive Scaling

- In this strategy, the system reacts immediately to changing demand of resources by adding or removing capacity on its own.
- System scales in response to the changing conditions based on some parameters.
- Reactive scaling approach should be seen as a safeguard for absolutely unavoidable scenarios.



Proactive vs. Reactive Scaling

- A scalable system should deal with varying workloads through proactive scaling approach as much as possible.
- Reactive scaling approach should be seen as a safeguard for absolutely unavoidable scenarios.
- Too much dependency on the reactive scaling strategy may turn suicidal.

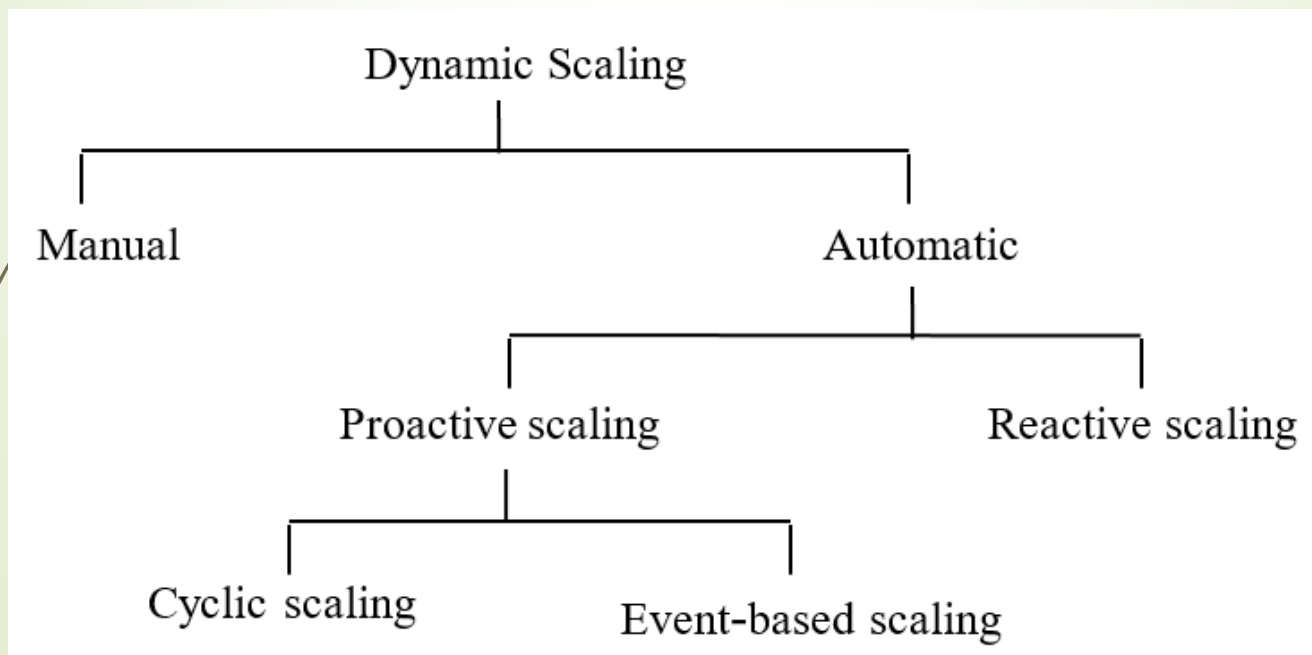


Proactive and Reactive Scaling

- **The Combination**
- A robust auto scaling implementation needs the appropriate mixture of two scaling approaches.
- In order to cope up with the real time scenarios, prediction-based proactive scaling is required to work together with reactive approach.

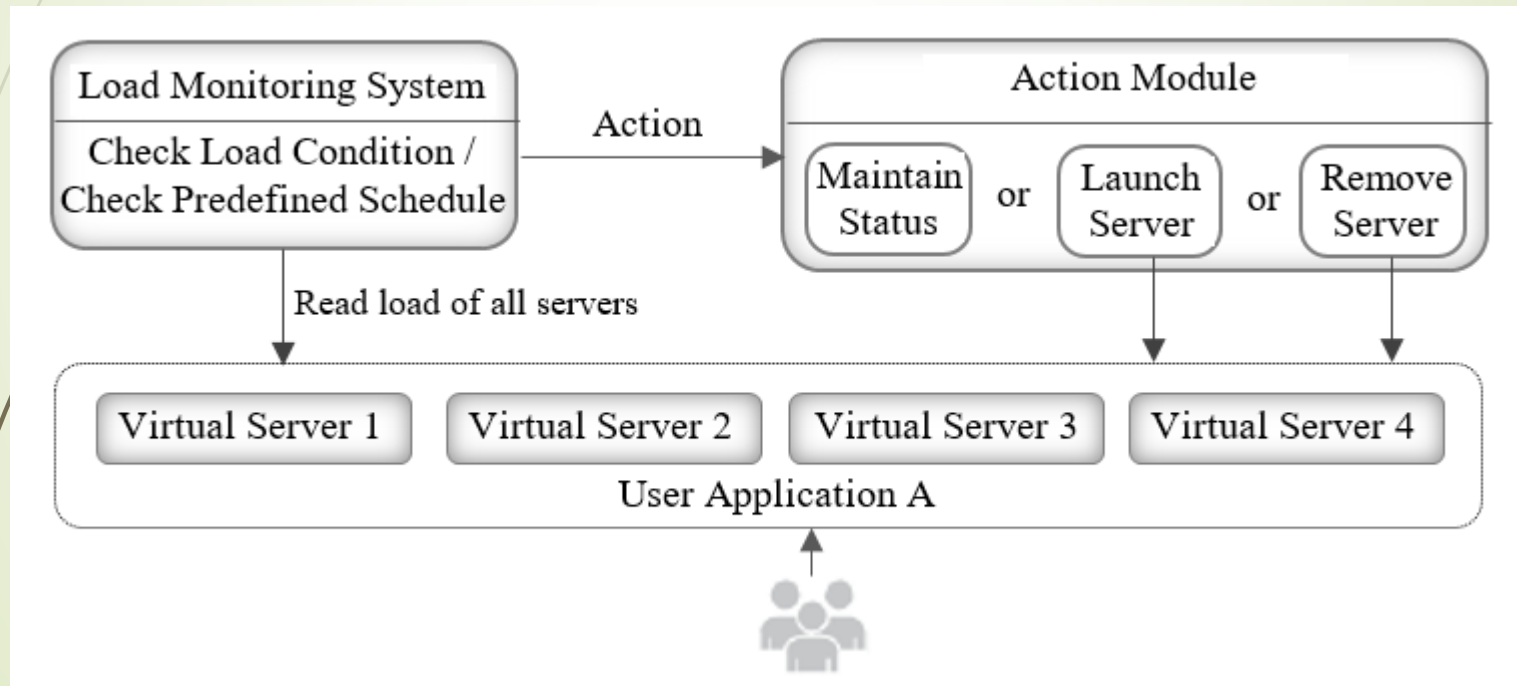
Dynamic Scaling Summary

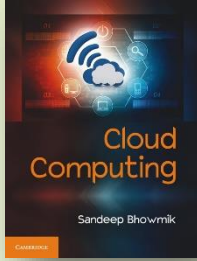
➤ Classification of dynamic scaling approaches



Auto Scaling in Cloud

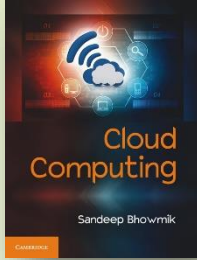
➤ The working model





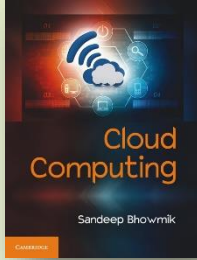
Scaling Boundaries

- Implementation of auto-scaling mechanism may not always mean giving up full manual control.
- To protect system from external attack, auto scaling mechanism is often limited by setting boundaries.
- Beyond those boundaries scaling is controlled manually.



Types of Scaling

- Scaling approaches can be categorized in two types –
 - Vertical Scaling
 - Horizontal Scaling

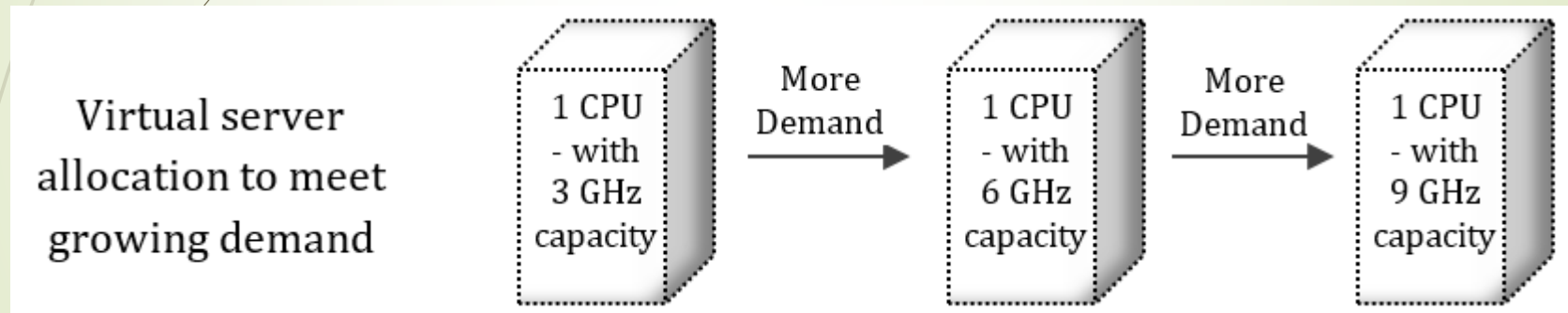


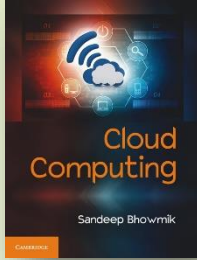
Vertical Scaling or Scaling Up

- Capacity can be increased by upgrading a resource quality to support raising workload.
- Similarly, capacity can be decreased by reducing a resource quality to support falling workload.
- This type of scaling where a resource component is powered up through replacement is called *scaling up* or *vertical scaling*.
- For example, a system with dual-processor capacity can be scaled vertically by replacing the processor with a quad-processor.

Vertical Scaling or Scaling Up

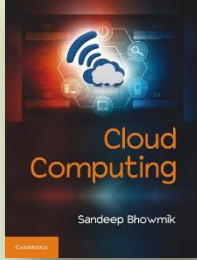
- Computing resource scaled up by replacing it with more powerful resource as demand increases





Vertical Scaling or Scaling Up

- **Advantage**
- It is not a complex task to replace component of a system, since that does not require alteration in system architecture.
- So, vertical scaling approach has low management complexity and less risky.
- **Disadvantage**
- High power hardware components or specialized resources are expensive.
- Capacity or power of any resource component always has an upper limit.

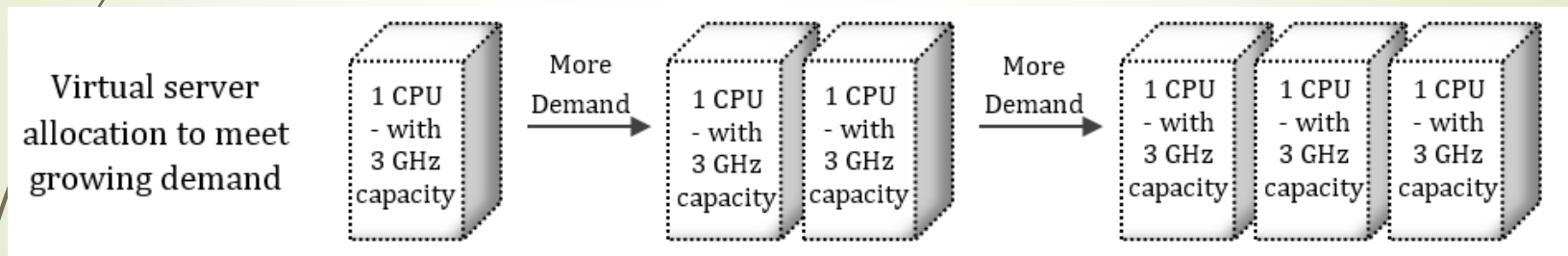


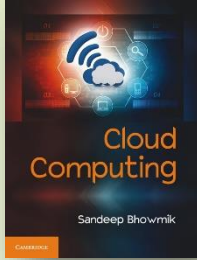
Horizontal Scaling or Scaling Out

- The overall system capacity is enhanced by adding more nodes.
- These new resources work together with the existing components.
- This type of scaling where system is equipped with supplementary resources to expand the resource capacity, is known as *scaling out* or *horizontal scaling*.
- For example, a system with dual-processor capacity can be scaled horizontally by adding another dual-processor.

Horizontal Scaling or Scaling Out

- Increase in system capacity through horizontal scaling is implemented by provisioning additional resource nodes.





Horizontal Scaling or Scaling Out

➤ Advantage

- Specialized high-end hardware components are not required, vastly available commodity hardware are used for horizontal scaling.
- This approach does not involve any system downtime.
- Unlike vertical scaling, no resource components are thrown out.

➤ Disadvantage

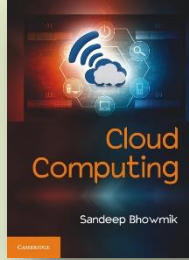
- Needs managing larger number of distributed nodes which synchronize together and hence involves complexity.

Comparison between Vertical and Horizontal Scaling

Vertical Scaling	Horizontal Scaling
Known as Scaling up.	Known as Scaling out.
This approach involves replacement of components or resource nodes.	This approach involves introduction of additional components or resource nodes.
It can be implemented in any type of computing environment.	It can only be implemented in distributed computing environment.
It has less management complexity.	The management of larger numbers of nodes increases system complexity.

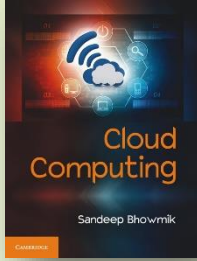
Comparison between Vertical and Horizontal Scaling

Vertical Scaling	Horizontal Scaling
May cause service interruption as system needs restart after replacing component.	Does not cause service interruption. No system restart required.
It has less influence on application architecture.	This approach has a more fundamental influence on application architecture.
A vertically scalable application tends to run on high end hardware.	A horizontally scalable application tends to run on low end (commodity) hardware.
Loads are concentrated.	Spreads the load.



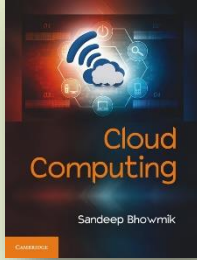
Comparison between Vertical and Horizontal Scaling

Vertical Scaling	Horizontal Scaling
Require specialized hardware components.	Can be done using normal commodity hardware components.
Upgrading the capacity of resource beyond a certain level can become very expensive.	Less expensive, and expense is always proportional with resource capacity.
Expansion is limited by any resource's maximum capacity.	Expansion does not depend on available capacity of hardware components.
Not a long term solution for scaling.	Provides a long term solution for scaling.



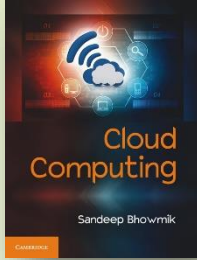
The Appropriate Choice

- The architectural structures of the vertical and horizontal scaling are different, and there are always tradeoffs between the two models.
- The decision depends on two aspects - the computational and business requirements.
- From computational viewpoint, the choice is linked with factors like the architecture of the system, design architecture of application.
- From business standpoint, availability of requirement, cost of implementation and system performance are the decisive factors.



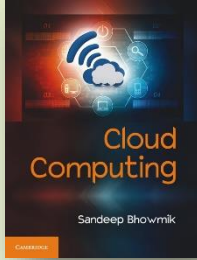
Horizontal Scaling is More Cloud-Native Approach

- Cloud computing model promises to deliver infinite scalability.
- Vertical scaling has limitations, and the infinite scalability promise can't be fulfilled with vertical scaling approach in cloud computing.
- Infinite scalability in cloud computing can only be achieved through horizontal scaling approach.
- Thus, horizontal scaling approach becomes more cloud native.



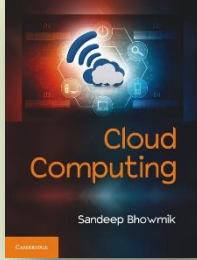
The Resource Contention Problem

- In computing, resource contention is about disputation over access to sufficient volume of resources.
- Often the *resource bottleneck problem* limits scalability.
- Different application has different stress points.



The Resource Contention Problem

- For cloud based applications the initial stress points are generally one among the following:
 - Process speed of application server
 - Memory capacity and speed of application server
 - Speed of the disk I/O operation of database server
 - Network bandwidth
- System designers must analyze the expected system load patterns and carry on tests to check how different scenarios create stress points in the infrastructure.

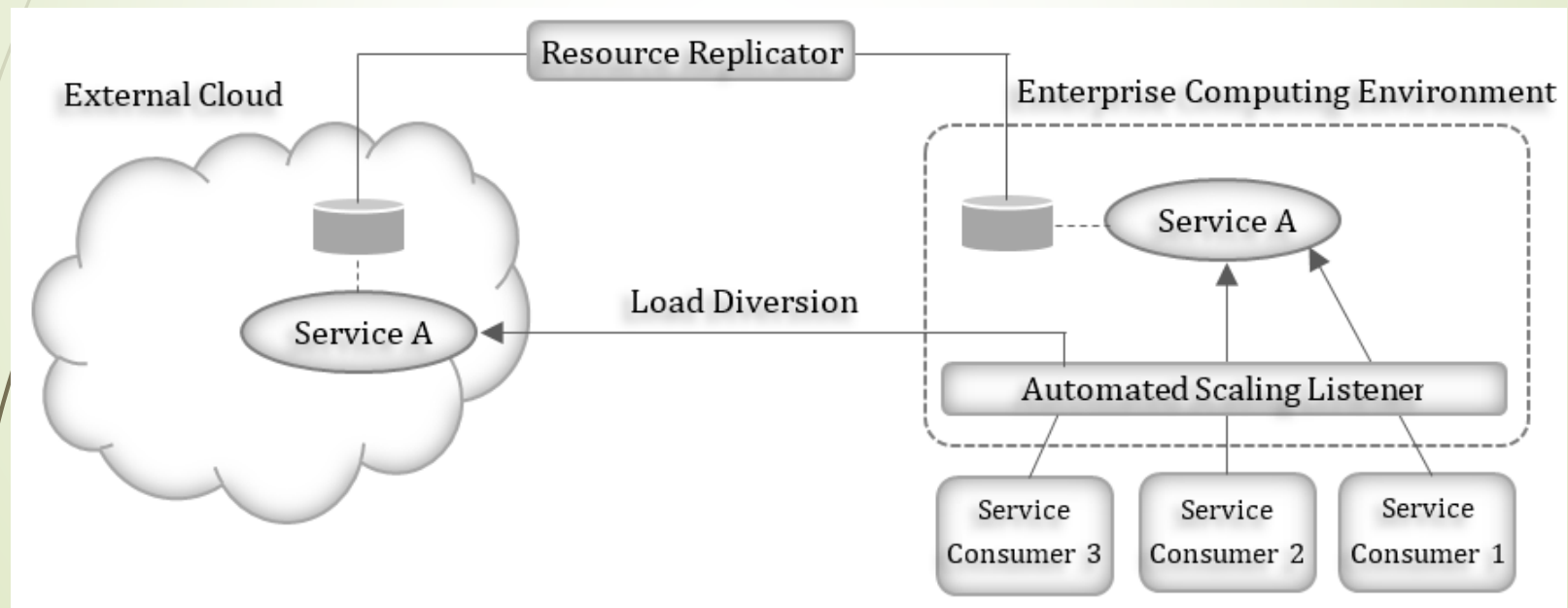


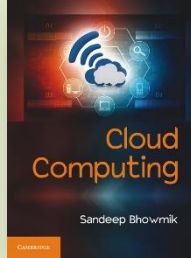
Cloud Bursting

- Cloudburst refers to the scaling where -
 - application primarily running on an organization's internal infrastructure or private cloud,
 - expands into some public cloud in the event of excessive
- At first resources of organization's internal computing system are consumed.
- With growing resource demand whenever a predefined capacity threshold is reached, the dynamic resource provisioning mechanism “bursts out”.

Cloud Bursting

➤ The Cloud Bursting Architecture





Thank You